

Multi-Step Prompt Workflow

RAG Hallucination Detector — Final-Year Project Pipeline

Project Overview

This project is designed to improve the reliability of Retrieval-Augmented Generation (RAG) systems. It accepts a user question, identifies the information needed, retrieves relevant passages from a document store, generates an answer using only the retrieved evidence, and then checks the answer for unsupported claims. The complete workflow is divided into clear stages so that each step can be tested, demonstrated, and explained during a project presentation or viva.

Step 1 — Query Understanding

Purpose: Understand the user's intent before searching the document collection.

You will receive a user question. Do not answer it yet.

1. Identify the main factual information requested.
2. Rewrite the request as a clear retrieval objective.
3. Extract 2-3 important search terms.

Question: "{user_query}"

Output:

CLAIM: ...

SEARCH_TERMS: ...

Step 2 — Evidence Retrieval

Purpose: Collect the evidence that will be used to answer the question.

Use the search terms from Step 1 to retrieve the top {k} relevant passages from the document index.

Return each passage with:

- Passage ID
- Similarity score
- Original passage text

Do not summarize, rewrite, or add information to the retrieved passages.

Step 3 — Evidence-Grounded Answer Generation

Purpose: Make the generated response traceable to the retrieved evidence.

Answer the user's question using ONLY the retrieved passages.

If the passages do not provide enough information, clearly state that the available evidence is insufficient.

Do not use unsupported information from general knowledge.

After each sentence, add the passage ID that supports it, for example: [P2].

{retrieved_passages}

Question: "{user_query}"

Answer:

Step 4 — Hallucination Detection

Purpose: Detect claims that are missing, exaggerated, or inconsistent with the retrieved sources.

Check every sentence of the draft answer against the retrieved passages.

For each sentence, classify it as:

- SUPPORTED – directly supported by the evidence
- PARTIALLY SUPPORTED – only part of the claim is supported
- UNSUPPORTED (HALLUCINATION) – the evidence does not support the claim

Give a short reason for every decision.

{draft_answer}

{retrieved_passages}

Step 5 — Confidence Scoring and Flagging

Purpose: Convert detailed sentence checks into an easy-to-understand answer-level result.

Given the sentence-level verdicts:

{verdict_list}

Calculate:

Overall confidence = (number of SUPPORTED sentences / total sentences) × 100

Set the result to:

FLAG = "REVIEW NEEDED" if any sentence is UNSUPPORTED.

FLAG = "OK" if no sentence is UNSUPPORTED.

Output:

SCORE: {score}%

FLAG: {flag}

Step 6 — Final Output and Reporting

Purpose: Provide transparency and create useful records for measuring the detector's performance.

Present the final result to the user or evaluator:

1. Display the generated answer.
2. Highlight unsupported or questionable sentences.
3. Show the confidence score and final flag.
4. Display the source passages and their IDs for verification.
5. Store the query, retrieved evidence, answer, verdicts, and score for later evaluation.

Workflow Summary

Stage	Process	Main Output
1	Query Understanding	Claim + search terms
2	Evidence Retrieval	Top-k relevant passages
3	Grounded Generation	Evidence-tagged draft answer
4	Hallucination Detection	Sentence-level verdicts
5	Confidence Scoring	Score percentage + flag
6	Final Reporting	Answer + evidence + evaluation log

How the Complete System Works

The system follows a controlled pipeline: the user question is first converted into a clear search objective. Relevant evidence is then retrieved from the document collection. The language model creates a response only from that evidence and attaches source identifiers to individual claims. A separate checking stage verifies whether each claim is actually supported. Finally, the system calculates a confidence score, flags risky answers for review, and shows the evidence used to produce the response.

Core Principle

An answer should reach the user only after its claims have been checked against retrieved evidence.

Unsupported claims should be clearly identified rather than hidden, making the RAG application more transparent, reliable, and easier to evaluate.

Expected Project Benefits

- Reduces unsupported or fabricated statements in RAG-generated answers.
- Makes every important answer claim traceable to retrieved evidence.
- Provides a simple confidence score and review flag for evaluators.
- Improves transparency by showing the passages used for generation.
- Creates structured logs that can later be used to evaluate detector accuracy.